

Ruturaj Marathe

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Summary

Software Engineer with **3+ years** experience specializing in transitioning ideas from research to enterprise products developing scalable software backend for automation, machine learning workflows and distributed systems.

Professional Experience

Visteon Corporation

Jun 2025 - Jan 2026

Software Engineer | Python (REST API, pywin32, M365), SQL Server, Siemens Mentor Graphics, Generative AI

Detroit, MI

- Designed and deployed FastAPI-based microservices to auto generate complex automotive test reports achieving a 85% process efficiency improvement.
- Integrated OCR and GPT-4o to extract unstructured data from block diagrams, auto generating the consolidated Bills of Materials, improving process efficiency by 70%.
- Implemented autonomous costing of Bill of Materials, reducing hardware architect's time by 50%.
- Engineered PL/SQL delta-load procedures to migrate the supplier finance data from Qlik to Microsoft Power BI, processing 5M+ records across 7+ reports, improving query performance and data governance.

Intel Corporation

May 2024 - Dec 2024

Software Engineer Intern | Python (pandas, model training-inference), Unix, HSDES, Pre-silicon validation

Santa Clara, CA

- Independently designed REST API-driven single click automation to create and track HSD bugs, cutting bug reporting time by 70%.
- Delivered a Machine Learning pipeline that runs DBSCAN clustering to group SoC design paths, auto generate waiver codes, leading to early detection of Setup/Hold violations, cutting down 2 design iterations.
- Developed Machine Learning solution on a central Unix server by training and inference of an XGBoost model to predict delays, reducing IO budgeting iterations from 10 to 6.
- Collaborated with cross-functional teams to develop a python script, reporting the Grey code Soc paths from the crossclock reports, cutting down STA time by 60%.

Visteon Corporation

Jan 2021 - Jul 2023

Software Engineer 2 | C/C++, Java, Git, Embedded Linux, ADB, SQLite

Bangalore, India

- Designed the favorites marking feature enabling fast-access to media playback, reducing access time by 80%.
- Built a fault-tolerant factory reset feature preserving media device state and achieving 100% reliability across device reboot.
- Debugged and resolved 45 defects across Java APIs, native middleware, and Linux system calls in a multi-threaded environment, enabling successful product release.

Skills

Languages	Python, Java, C/C++, SQL
Frameworks & Libraries	FastAPI, Pandas, NumPy, Scikit-learn, Android Multimedia
Cloud & Infrastructure	Microsoft (Azure OpenAI, AD, Cost monitor), Power Automate, AWS, GCP, Docker, Git
Data & Analytics	Oracle, SQL Server, Redis, Power BI, Data modeling
Systems & Architecture	Linux, Multithreading, Distributed Systems, RESTful APIs, Microservices

Projects

DoctorAssist.ai | Generative AI, NLP, Gemini API, FAISS

- Developed a tool enabling natural-language querying over relational healthcare data, translating natural language queries into SQL and improving data access speed by 40%.

Used Car Data Analysis | AWS, EC2, S3, boto3, PySpark

- Built a distributed data pipeline to generate pricing insights from car listings, providing sales enhancement strategy.

Education

Indiana University Bloomington, IN, United States

Graduated

M.S. in Computer Science,

Aug 2023 - May 2025

Coursework: Software Engineering, Applied Machine Learning, System Design, Algorithms

Visvesvaraya Technological University, KA, India

Graduated

B.E in Computer Science

Aug 2016 - Jun 2020

Coursework: Data Structures, Object-Oriented Programming, Operating Systems